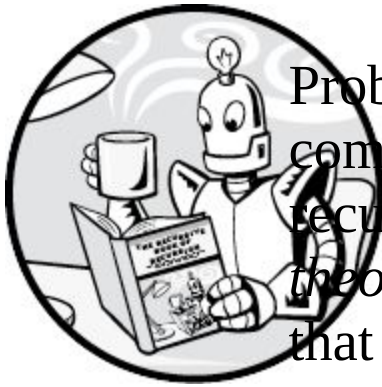


6

PERMUTATIONS AND COMBINATIONS



Problems involving permutations and combinations are especially suited to recursion. These are common in *set theory*, a branch of mathematical logic that deals with the selection, arrangement, and manipulation of collections of objects.

Dealing with small sets in our short-term memory is simple. We can easily come up with every possible order (that is, *permutation*) or combination of a set of three or four objects. Ordering and combining items in a larger set requires the same process but quickly turns into an impossible task for our human brains. At that point, it becomes practical to bring in computers to handle the combinatorial explosion that occurs as we add more objects to a set.

At its heart, calculating permutations and combinations of large groups involves calculating permutations and combinations of smaller groups. This makes these calculations suitable for recursion. In this chapter, we'll look at recursive algorithms for generating all possible permutations and combinations of characters in a string. We'll expand on this to generate all possible combinations of balanced parentheses (orderings of open parentheses correctly matched to closing parentheses). And finally, we will calculate the power set of a set—that is, the set of all possible subsets of a set.